

Fangshuo (Jasper) Liao

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RESEARCH INTEREST	I am primarily interested in a scientific and theoretical understanding of various aspects in machine learning, including optimization, representation, generalization, and scaling behaviors. I am also interested in applying such understanding to development of practical approaches.	
ACADEMIC APPOINTMENT	Assistant Professor/Faculty Fellow Courant Computer Science, New York University	2026-now
EDUCATION	Ph.D. Computer Science George R. Brown School of Engineering, Rice University Advisor: Prof. Anastasios Kyrillidis [website]	2020-2026 GPA: 4.00/4.00
	B.S. Computer Science George R. Brown School of Engineering, Rice University	2016-2020 GPA: 3.92/4.00
	B.A. Mathematics Wiess School of Natural Science, Rice University	2016-2020 GPA: 3.92/4.00
OTHER RESEARCH EXPERIENCE	Baylor College of Medicine <i>Undergraduate research assistant working with Prof. Robert Waterland</i> Finding genetic sequence blocks with systematic individual variation in epigenetics.	Jun.2018-Sept.2018
CONFERENCE PAPER	Yehya Farhat, Hamza ElMokhtar Shili, Fangshuo Liao , Chen Dun, Mirian Hipolito Garcia, Guoqing Zheng, Ahmed Hassan Awadallah, Robert Sim, Dimitrios Dimitriadis, Anastasios Kyrillidis. <i>DDOME: Joint Gating-Expert Training for Adaptive MoEs in Decentralized Settings</i> , NeurIPS, 2025. [Link] Fangshuo Liao , Wenyi Su, and Anastasios Kyrillidis, "Provable Model-Parallel Distributed Principal Component Analysis with Parallel Deflation", CPAL, 2025. [Link] Fangshuo Liao , Junhyung Lyle Kim, Cruz Barnum, and Anastasios Kyrillidis, "On the Error-Propagation of Inexact Deflation for Principal Component Analysis", ICML, 2024. [Link] Fangshuo Liao and Anastasios Kyrillidis, "Provable Accelerated Convergence of Nesterov's Momentum for Deep Neural Networks", ALT, 2024. [Link] Zichang Liu, Aditya Desai, Fangshuo Liao , Weitao Wang, Victor Xie, Zhaozhuo Xu, Anastasios Kyrillidis, Anshumali Shrivastava, "Scissorhands: Exploiting the Persistence of Importance Hypothesis for LLM KV Cache Compression at Test Time", NeurIPS, 2023. [Link] Zheyang Xiong*, Fangshuo Liao * and Anastasios Kyrillidis, "Strong Lottery Ticket Hypothe-	

*Equal Contribution

sis with ϵ -perturbation", AISTATS, 2023. [Link]

Qihan Wang*, Chen Dun*, **Fangshuo Liao*** and Anastasios Kyrillidis, "LOFT: Finding Lottery Tickets through Filter-wise Training", AISTATS, 2023. [Link]

JOURNAL PAPER

Cameron R Wolfe*, **Fangshuo Liao***, Qihan Wang, Junhyung Lyle Kim, Anastasios Kyrillidis, "How Much Pre-training Is Enough to Discover a Good Subnetwork?", Transactions on Machine Learning Research (TMLR), 2024. [Link]

Fangshuo Liao and Anastasios Kyrillidis, "On the Convergence of Shallow Neural Network Training with Randomly Masked Neurons", Transactions on Machine Learning Research (TMLR), 2022. [Link]

Cameron R Wolfe*, Jingkang Yang*, **Fangshuo Liao***, Arindam Chowdhury, Chen Dun, Artun Bayer, Santiago Segarra, Anastasios Kyrillidis, "GIST: Distributed Training for Large-Scale Graph Convolutional Networks", Journal of Applied and Computational Topology, 2023. [Link]

PREPRINT

Fangshuo Liao, Anastasios Kyrillidis, *Guided by the Experts: Provable Feature Learning Dynamic of Soft-Routed Mixture-of-Experts*, under review, 2025. [Link]

Mahtab Alizadeh Vandchali*, **Fangshuo Liao***, Nikhil Chigali, Anastasios Kyrillidis, "One Rank at a Time: Cascading Error Dynamics in Sequential Learning", under review, 2025. [Link]

TEACHING

CSCI-UA.0473-002/003 – Fundamentals of Machine Learning - Recitation

- Students enrolled: 150 (Fall 2026)
- Instructor evaluation: not available yet.

TEACHING ASSISTANT

COMP 282 – Computational Optimization for AI

- Spring 2026, head TA
- Helping design course material, organizing TA team, giving lecture on specific topics, grading homework, and holding office hours.

COMP 540 – Statistical Machine Learning

- Spring 2022, 2021, 2020
- Designing course projects, improving and grading homework, giving multiple recitation lectures, and holding office hours.

COMP 440/557 – Artificial Intelligence

- Fall 2021, 2019
- Improving and grading homework, giving recitation lectures, holding office hours.

MENTORSHIP

With Prof. Anastasios Kyrillidis

- **Barbara (Wenyi) Su** (Rice University) Feb.2024-May.2026
Distributed Principal Component Analysis with Parallel Deflation.
- **Aaron Duong & Albert Zhu** (Rice University) May.2023-May.2024
Efficient Distributed Linear Regression via Feature Subsampling.
- **Isabel Cevallos** (Villanova University) May.2023-Aug.2023
Distributed Principal Component Analysis with Deflation Method.
- **Zheyang (Eddie) Xiong** (Rice University) Aug.2021-May.2023
Strong Lottery Ticket Hypothesis with ϵ -Perturbation.
- **Yuan Gao** (Purdue University) May.2022-Aug.2022
Federated Learning using Graph Independent Subnet Training.

– **Kaichun Luo** (Rice University)

May.2020-Aug.2021

Sparse Simplex Projection for Multi-label Classification and Neural Architecture Search.

**INVITED TALKS
& WORKSHOPS**

Provable Accelerated Convergence of Nesterov’s Momentum for Deep Neural Networks. ICCOPT. July, 2025.

Understanding and Utilizing Error Propagation in Hotelling’s Deflation for PCA. TL;DR. May, 2025.

Provable Model-Parallel Distributed Principal Component Analysis with Parallel Deflation. CPAL (Poster). March, 2025.

Strong Lottery Ticket Hypothesis with ϵ -perturbation. NeurIPS OPT-ML Workshop (Oral). December, 2022.

LoFT: Finding Lottery Tickets through Filter-wise Training. NeurIPS HITY Workshop (Poster). December, 2022.

GIST: Distributed Training for Large-Scale Graph Convolutional Networks. NeurIPS GLFrontier Workshop (Poster). December, 2022.

LoFT: Finding Lottery Tickets through Filter-wise Training. Intel’s MLWiNS Annual Workshop. October, 2023.

LoFT: Finding Lottery Tickets through Filter-wise Training. Intel’s MLWiNS Annual Workshop. October, 2022.

Provable distributed Learning of Deep Neural Networks using Independent Subnet Training. Intel’s MLWiNS Mid-Year Workshop. April, 2022.

On the Convergence of Shallow Neural Network Training with Randomly Masked Neurons. Google’s Federated Learning and Analytics Workshop. November, 2021.

SERVICE

Reviewer:

– AISTATS 2023,2026; ICML 2023-2025; ICLR 2024-2026, NeurIPS 2024-2025, TMLR.

Workshop:

– TL;DR 2023,2025: Co-organizer for “Texas Colloquium on Distributed Learning” [Website]